Example 9.33

Evaluate : $\int_0^{2\pi} x^2 \sin nx\, dx$, where n is a positive integer.

Solution

Taking $u = x^2$ and $v = \sin nx$, and applying the Bernoulli's formula, we get

$$I = \int_0^{2\pi} x^2 \sin nx\, dx = \left[\left(x^2\right)\left(-\frac{\cos nx}{n}\right) - (2x)\left(-\frac{\sin nx}{n^2}\right) + (2)\left(\frac{\cos nx}{n^3}\right)\right]_0^{2\pi}$$

$$= \left[\left(4\pi^2\right)\left(-\frac{1}{n}\right) - 0 + (2)\left(\frac{1}{n^3}\right)\right] - \left[0 - 0 + (2)\left(\frac{1}{n^3}\right)\right], \text{ since } \cos 2n\pi = 1 \text{ and } \sin 2n\pi = 0$$

$$= -\frac{4\pi^2}{n} + \frac{2}{n^3} - \frac{2}{n^3} = -\frac{4\pi^2}{n}.$$

Example 9.34

Evaluate : $\int_{-1}^{1} \mathrm{e}^{-\lambda x}(1 - x^2)\, dx$.

Solution

Taking $u = 1 - x^2$ and $v = \mathrm{e}^{-\lambda x}$, and applying the Bernoulli's formula, we get

$$I = \int_{-1}^{1} \mathrm{e}^{-\lambda x}(1 - x^2)\, dx = \left[(1 - x^2)\left(\frac{e^{-\lambda x}}{-\lambda}\right) - (-2x)\left(\frac{e^{-\lambda x}}{\lambda^2}\right) + (-2)\left(\frac{e^{-\lambda x}}{-\lambda^3}\right)\right]_{-1}^{1}$$

$$= 2\left(\frac{e^{-\lambda}}{\lambda^2}\right) + 2\left(\frac{e^{-\lambda}}{\lambda^3}\right) + 2\left(\frac{e^{\lambda}}{\lambda^2}\right) - 2\left(\frac{e^{\lambda}}{\lambda^3}\right)$$

$$= \frac{2}{\lambda^2}\left(e^{\lambda} + e^{-\lambda}\right) - \frac{2}{\lambda^3}\left(e^{\lambda} - e^{-\lambda}\right).$$

EXERCISE 9.4

Evaluate the following:

1. $\int_0^1 x^3 e^{-2x}\, dx$
2. $\int_0^1 \frac{\sin(3\tan^{-1} x)\tan^{-1} x}{1 + x^2}\, dx$
3. $\int_0^{\frac{1}{\sqrt{2}}} \frac{e^{\sin^{-1} x} \sin^{-1} x}{\sqrt{1 - x^2}}\, dx$
4. $\int_0^{\frac{\pi}{2}} x^2 \cos 2x\, dx$

9.5 Improper Integrals

In defining the Riemann integral $\int_a^b f(x)dx$, the interval $[a,b]$ of integration is finite and $f(x)$ is finite at every point in $[a,b]$. In many physical applications, the following types of integrals arise:

$$\int_a^{\infty} f(x)\, dx,\ \int_{-\infty}^{a} f(x)\, dx,\ \int_{-\infty}^{\infty} f(x)\, dx,$$

where a is a real number and $f(x)$ is a continuous function on the interval of integration. They are defined as the limits of Riemann integrals as follows:

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